Berry Holonomy and Color Holonomy in the Rope Framework

Geometric-Phase Structure of Rope Transport — What Is Computed and What Is Analogy

Mark Palmer · with computational collaboration by Claude (Anthropic)

Global Networking Practice, NTT DATA, Charlotte, NC · palmer100@gmail.com

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it.

Abstract

This paper reports the geometric-phase (holonomy) structure that arises when rope configurations are transported around closed loops in their parameter space. The two-strand Hopf geometry supports a Berry-type holonomy, and the framework draws an analogy to colour holonomy in the non-Abelian case. We separate carefully what is COMPUTED from the Hopf geometry (the Berry phase structure) from what is offered as structural ANALOGY (the colour case), and label each accordingly. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

> **Scope of this paper**
>
> CLAIM: the two-strand Hopf geometry supports a Berry-type geometric phase under closed-loop transport.
>
> NOT CLAIMED: a full non-Abelian colour-holonomy derivation — that is offered as structural analogy, not computed. Status: Modeled (Berry) / Open (colour).

1. Berry Holonomy From the Hopf Geometry

Transporting a two-strand rope state around a closed loop in its parameter space accumulates a geometric phase determined by the Hopf bundle's connection — a Berry phase. This structure follows from the same Hopf geometry that appears throughout the rope treatment of the two-strand state, and is a genuine geometric feature of the model.

2. Colour Holonomy: Analogy, Not Derivation

Marked clearly as analogy. The extension to non-Abelian 'colour' holonomy is presented as a structural analogy to the Berry case, NOT as a computed result. Establishing a genuine colour holonomy would require the non-Abelian connection to be derived from the rope dynamics, which is not done here. This is flagged as open so the analogy is not mistaken for a derivation.

> **Status**
>
> Berry geometric phase from Hopf transport — Modeled (geometric feature).

Colour (non-Abelian) holonomy — Open (offered as analogy, not derived).

Programme credit: the core Rope Hypothesis is due to Bill Gaede. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.